

```
align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[subtitle] linebreak() text(fill: black70, size: 6.8pt)[Five HRF images across healthy, diabetic-
retinopathy, and glaucoma cases. Filtering uses the green channel only. Noisy metrics average over
100 AWGN draws.] ] v(8pt) image-grid(data, [Gradient magnitude, clean HRF retinal images],
"magnitude_path", show-input: true)
```

```
align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Orientation maps. Hue encodes edge angle modulo  $\pi$  and brightness follows normalized
magnitude.] ] v(8pt) image-grid(data, [Gradient orientation, clean HRF retinal images],
"orientation_path", show-input: false)
```

```

align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Quantitative summary on vessel-boundary ODS, boundary-vector RMSE, and centerline tangent
orientation MAE.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, [ metric-chart(data, "ods_f_score",
[Soft-label ODS F-score versus SNR], [ODS F-score]) ], [ metric-chart(data,
"gradient_vector_rmse_mean", [Gradient-vector RMSE versus SNR], [Vector RMSE]) ], [ metric-
chart(data, "orientation_mae_deg_mean", [Centerline tangent MAE versus SNR], [Angle MAE
(deg)]) ], [ summary-table(data) ], ) v(10pt) legend(data)

```

```

align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[WVF radius-trace follow-up on the fixed HRF image set.] linebreak() text(fill: black70, size: 6.8pt)
[The garnet trace sweeps WVF across  $(r, d) = (3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11)$ . Thin
horizontal lines are the fixed baseline methods.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, row-
gutter: 10pt, [ wvf-trace-chart(data, "gradient_vector_rmse_mean", "inf", [Vector RMSE versus WVF
radius, clean], [Vector RMSE]) ], [ wvf-trace-chart(data, "orientation_mae_deg_mean", "inf",
[Orientation MAE versus WVF radius, clean], [Angle MAE (deg)]) ], [ wvf-trace-chart(data,
"gradient_vector_rmse_mean", "10", [Vector RMSE versus WVF radius, 10 dB], [Vector RMSE]) ],
[ wvf-trace-chart(data, "orientation_mae_deg_mean", "10", [Orientation MAE versus WVF radius, 10
dB], [Angle MAE (deg)]) ], ) v(8pt) legend(data)

```

```

align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Best-WVF versus best-small-stencil deltas. Positive values mean the best WVF grid cell has lower
error than the best 3×3 stencil baseline.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, [ delta-
chart(data, "gradient_vector_rmse_mean", [Vector RMSE delta, small-stencil minus best WVF],
[RMSE delta]) ], [ delta-chart(data, "orientation_mae_deg_mean", [Orientation MAE delta, small-
stencil minus best WVF], [Angle delta (deg)]) ], ) v(10pt) delta-legend()

```